PulseOS

Smart Health Monitoring System · B.Tech CSE Major Project

Health alerts that learn the patient

Self-learning, explainable alerts with patient and doctor feedback, powered by our AdaptAlert engine

~97%
fewer false alarms

100%
real events caught

7
stage explainable AI

3
apps: patient, doctor, nurse

The problem: too many false alarms

Health monitors alarm whenever a number crosses one fixed line, for example heart rate above 100.

At home that line is crossed all the time by harmless things. Most alarms are false, so people start ignoring them. This is **alarm fatigue**, a recognised patient-safety problem.



A walk

Heart rate goes above 100 → fixed monitor alarms



A strong coffee

Heart rate stays high for minutes → fixed monitor alarms



A loose band

Readings jump all over the place → fixed monitor alarms

In our 30-day simulation of a fixed-threshold monitor

3.7

false alarms per patient, every day

3%

of its alerts were real

Source: The Joint Commission, Sentinel Event Alert 50 (2013)

Why existing monitoring systems fall short

One threshold for everyone

A fit 24-year-old and a 70-year-old heart patient get the same alarm line.

They never learn

The same harmless pattern alarms again and again, even after a doctor says it is fine.

Black-box alerts

An alert says “abnormal” but not why, so doctors and patients don’t trust it.

Bad sensor data

A loose band or movement looks like a medical problem.

Our solution: PulseOS with AdaptAlert

Learns each patient’s normal

For each activity: resting, light activity, active, cooling down

Explains every alert

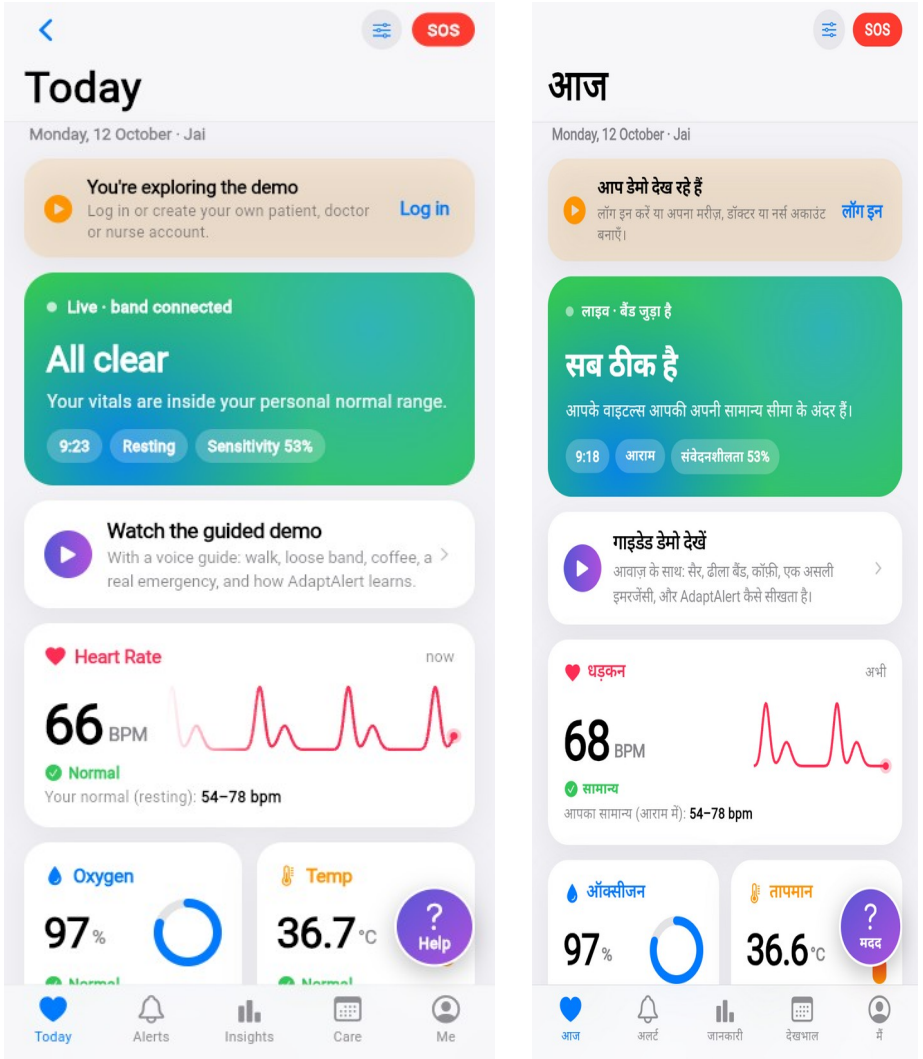
Plain-language reasons, how sure it is, and what would change the decision

Learns from feedback

“I was walking” from the patient, “false alarm” or “real” from the doctor

Never ignores danger

A safety floor and bounded learning keep critical readings alerting



How AdaptAlert decides: a 7-stage pipeline

Every minute the wrist band sends heart rate, oxygen (SpO_2), temperature and movement. Each reading goes through:

A
Signal quality
Loose band or motion? The reading is ignored

B
Activity
Resting, light, active or cooling down

C
Personal normal
Average and spread for this person and activity

D
Safety floor
e.g. $SpO_2 \leq 90\%$ → critical alert, no AI needed

E
Anomaly
Unusual for this person for 3+ minutes

F
Decide
Model + this person's history vs threshold τ

G
Explain
Reasons, confidence, what would change it

Key idea: the safety floor (D) works without the AI, so learning can never switch off a critical alert.

The learning rule

$$P(real) = (1-w)\cdot P_{model} + w\cdot P_{you} \quad \cdot \quad w = n/(n+4) \quad \cdot \quad \text{alert if } P(real) \geq \tau$$

P_{model}

Decision tree built from clinical reasoning: how many vitals are abnormal, how far from normal, activity, time of day.

P_{you}

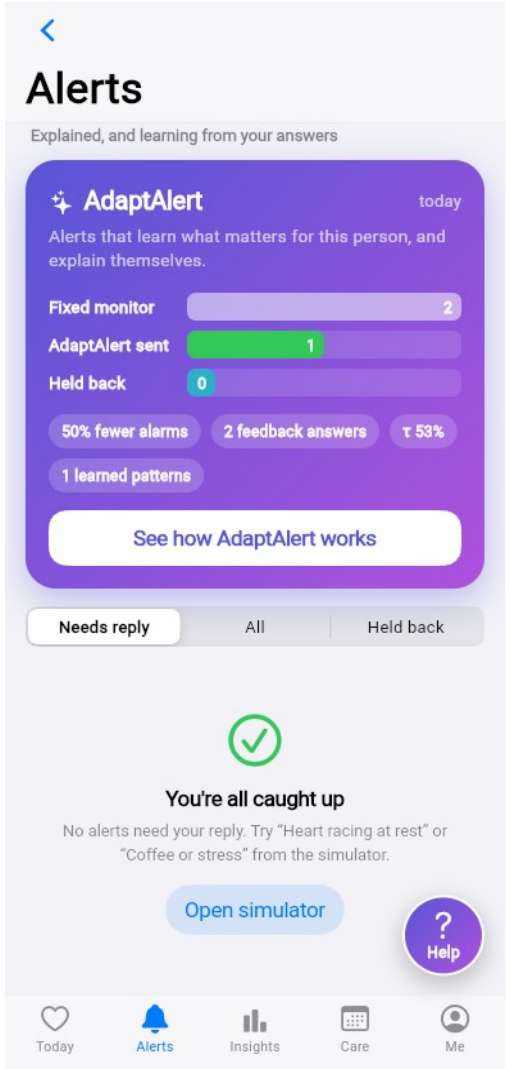
How often this same pattern was really a problem for this patient, from their and the doctor's answers. More answers (n) → more weight.

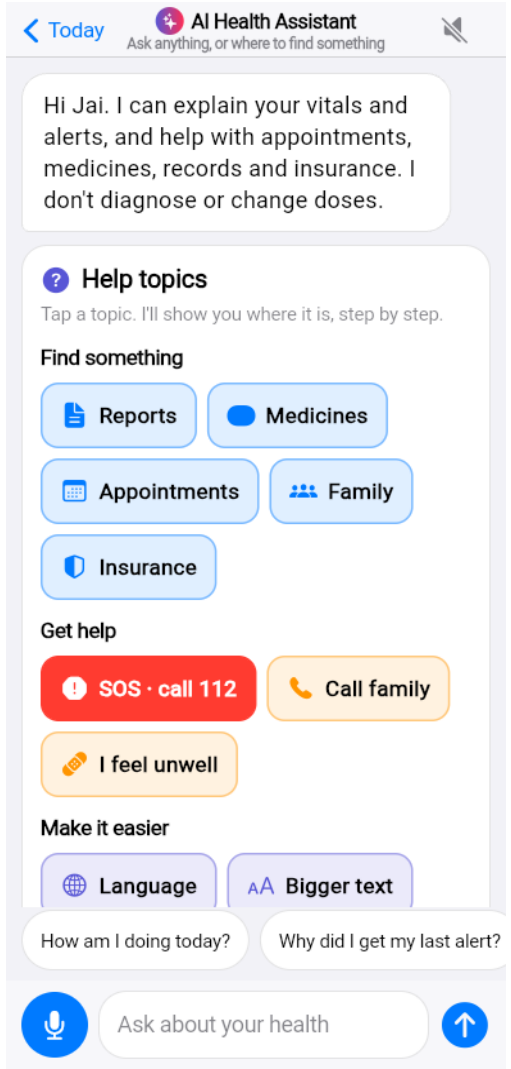
τ (sensitivity)

Personal threshold. "False alarm" from the doctor raises it a little; "real" lowers it more. Kept between 30% and 75%, moves at most 10 points a week.

Alerts that fall below τ are held back, not deleted: the patient and doctor can still see them, and anything worse alerts at once.

Every alert explains itself





Why

"Heart rate 118, which is 3.1 SD above your resting normal, while you were resting."

How sure

A likelihood with a 90% range, shown as model vs your history.

What would change it

"No alert below 104 bpm." "If only heart rate had changed, it would be rated lower."

Who was told

A timeline: patient, caregiver, doctor, SOS.

Feedback

Patient: I was active / sensor loose / I feel unwell. Doctor: real or false alarm.

Research gaps we address

G1 **One threshold for all**

→ Personal normal per vital and activity

G2 **No learning after deployment**

→ Learns from patient and doctor feedback

G3 **Learning can hide emergencies**

→ Safety floor + bounded, slow learning

G4 **Black-box alerts**

→ Reasons, confidence, counterfactuals

G5 **Bad sensor data**

→ Signal-quality check before anything else

G6 **Cold start for new patients**

→ Federated prior: shared label statistics, no raw data

G7 **No uncertainty in results**

→ 5 systems, bootstrap CIs, ablation, fatigue sweep

G8 **English-only apps**

→ Full English and Hindi, Hindi voice and assistant

G9 **Hard for elderly or disabled users**

→ Large text, easy home, flash and spoken alerts

How we tested it

40
virtual patients

30
days each

159
real deteriorations

5
systems compared

Every day each patient has exercise, a loose sensor, coffee or stress, night oxygen dips, and sometimes a real problem (fast heart with falling oxygen, fever, critical oxygen drop). All five systems see exactly the same events.

A
Fixed threshold
today's monitors

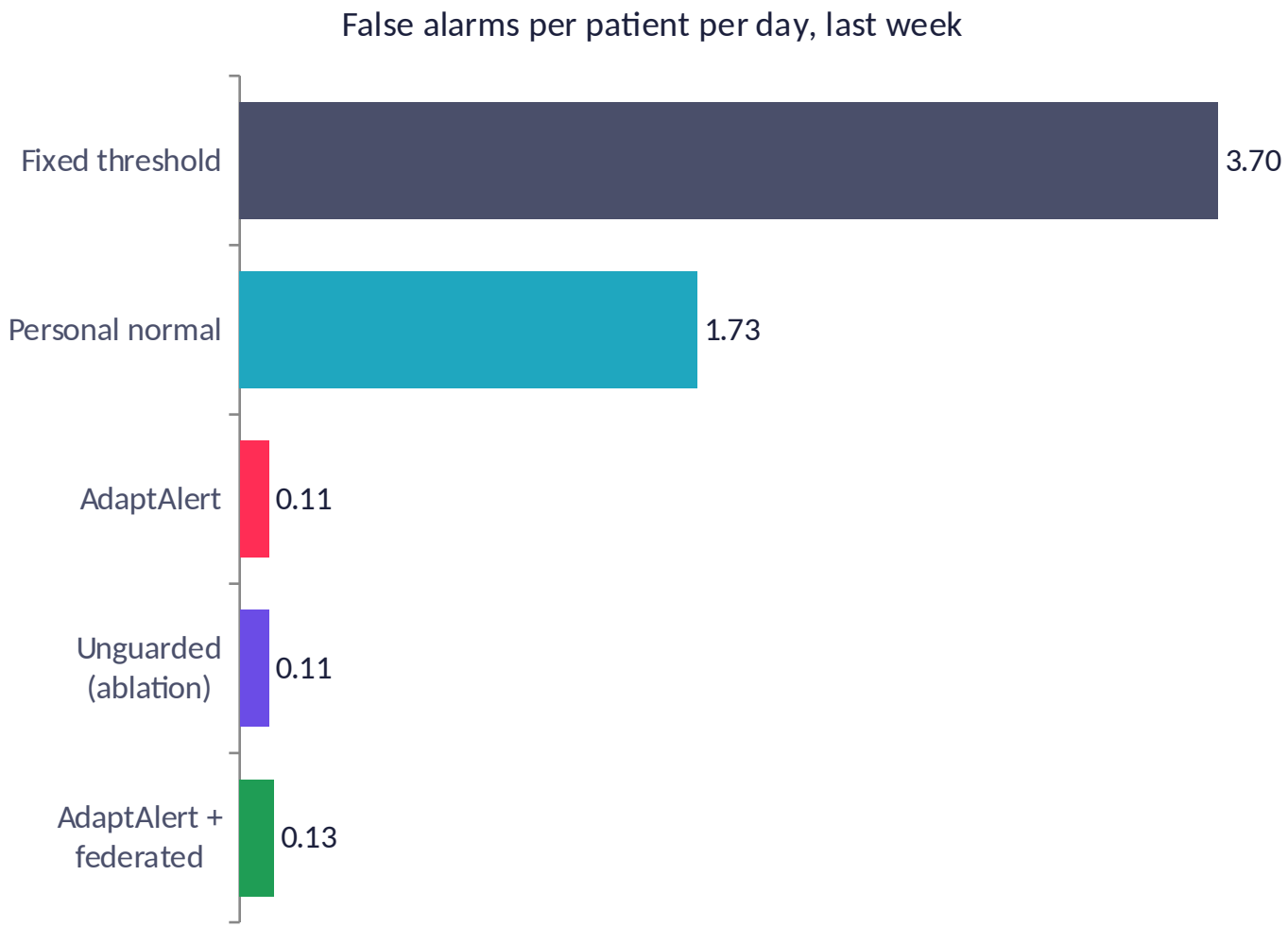
B
Personal normal
no learning

C
AdaptAlert
our system

D
Unguarded learning
ablation: no safeguards

E
AdaptAlert + federated
shared prior for new patients

Results: about 97% fewer false alarms



96.9%

fewer false alarms than fixed thresholds

100%

real events and critical events caught

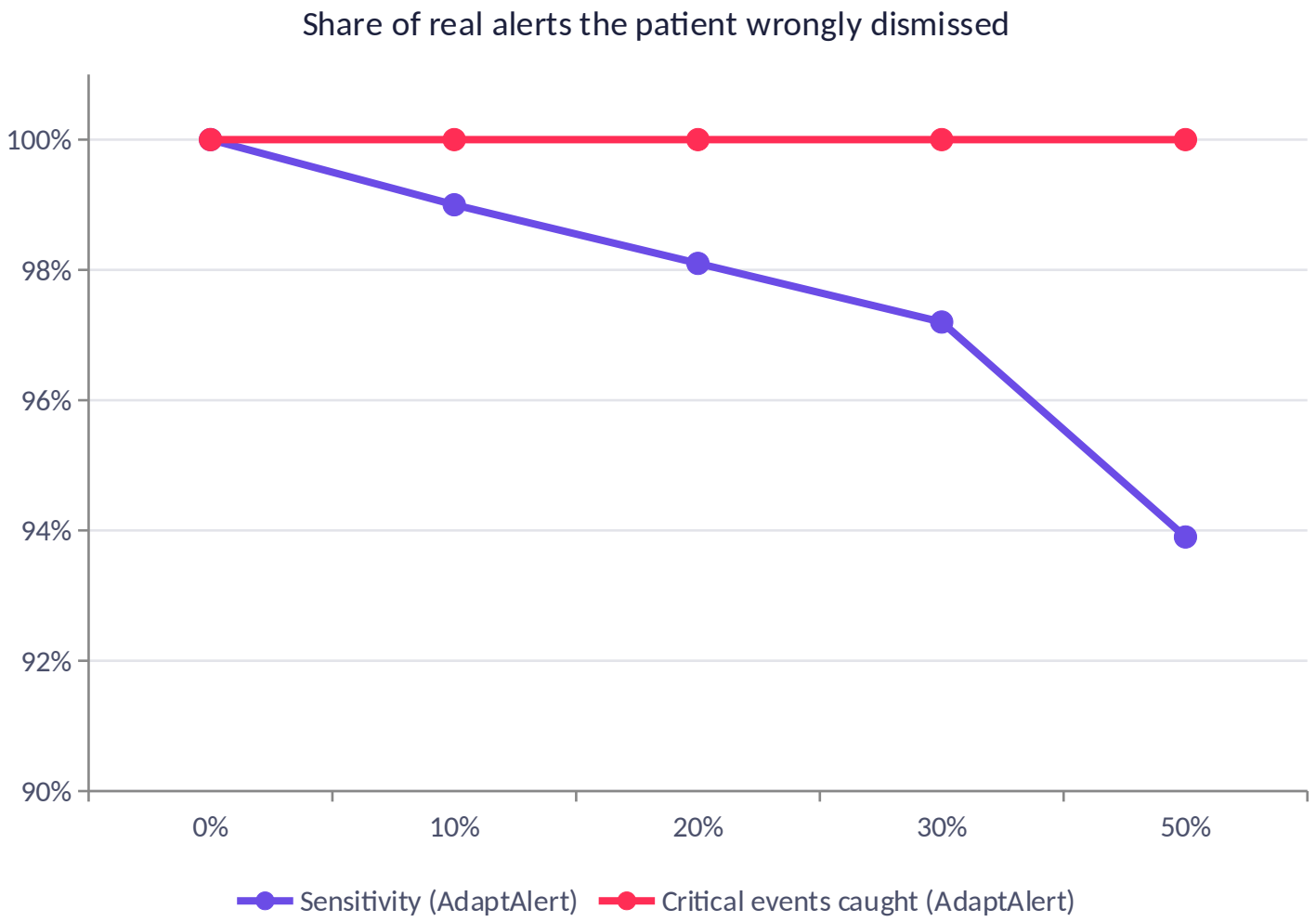
40% vs 3%

of alerts were real (precision)

0.47 → 0.06

week-1 false alarms with the federated prior

Safe even when people dismiss real alerts



What if patients press “false alarm” on real alerts?

- Even when half of the real alerts are dismissed, sensitivity stays near 94%.
- Critical events stay at 100%, because the safety floor does not use the learned model.
- Doctor labels outweigh patient answers, which limits the damage.

One website, three apps

Patient

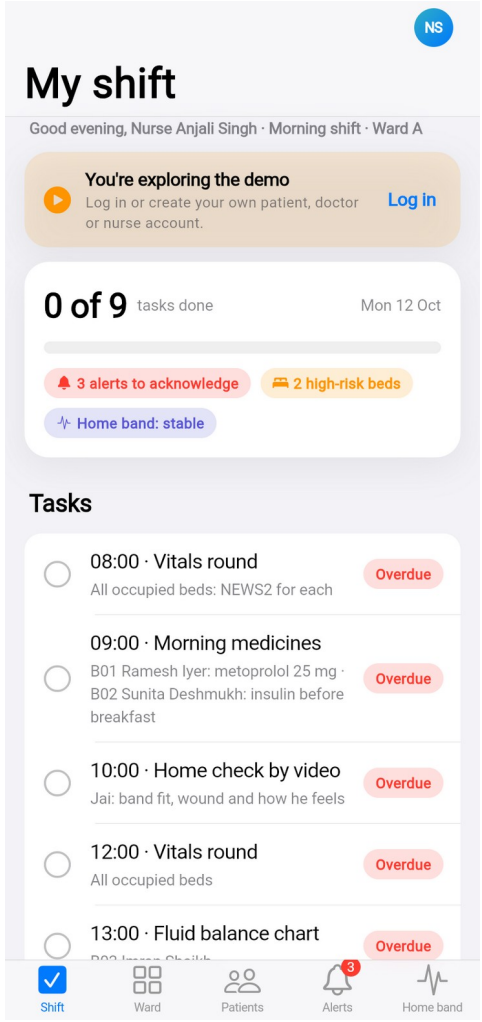
- Live vitals and status
- Explained alerts and feedback
- Appointments and doctors
- Records, insurance, family
- SOS and Medical ID

Doctor

- Today dashboard
- Patients: notes, prescriptions
- Schedule and follow-ups
- Review AdaptAlert alerts
- Ward with NEWS2 scores

Nurse

- Shift tasks with times
- Handover notes
- Ward alerts: acknowledge
- Home patient band status
- Patient notes



Built for everyone

English and हिन्दी

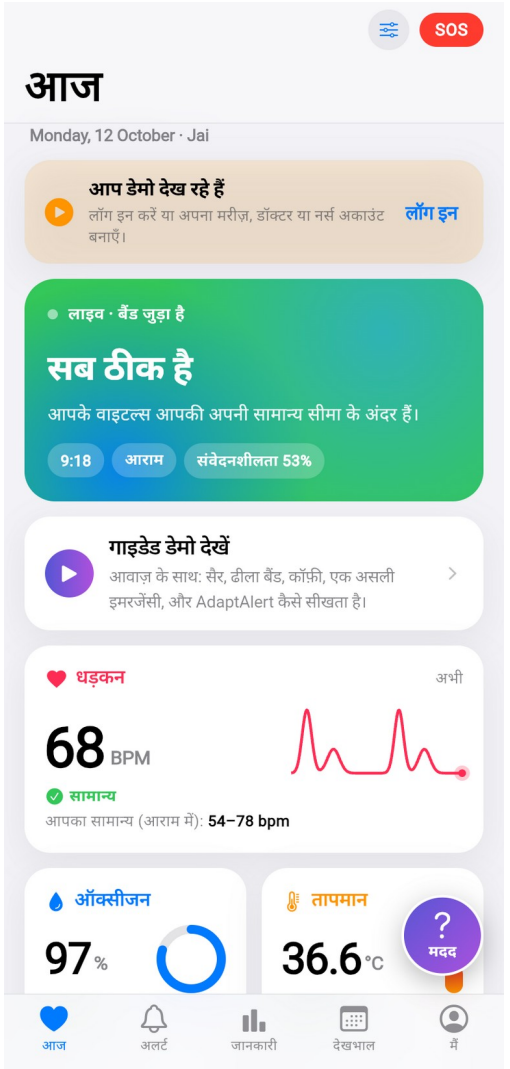
Every screen, the alert explanations, the assistant and the demo voice-over.

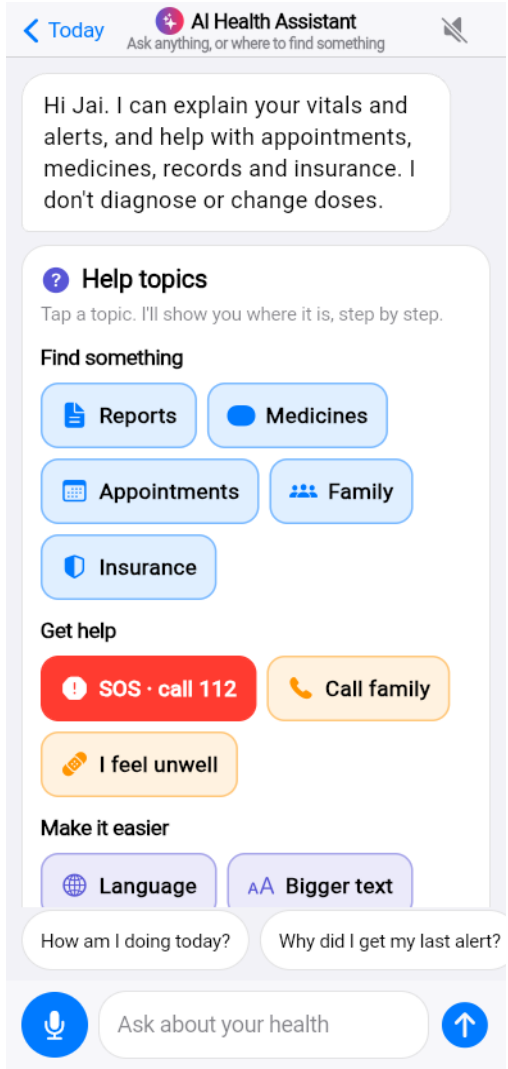
Help assistant

“Report kahan hai?” → steps + a button that goes there. Speak questions; hear answers.

Accessibility

Large text, easy home for older users, flash alerts for hearing loss, spoken alerts for low vision.





How it is built

Wrist band

Simulated now; ESP32 (MAX30102, MLX90614, MPU6050) or Apple Watch next

AdaptAlert engine

Pure Dart, runs on the device: 7 stages, learning, explanations

PulseOS apps

Flutter web, Android, iOS: patient, doctor, nurse

Research Lab

Runs the 5-system experiment in the app; copy results for the paper

Data

On-device accounts (salted SHA-256), optional Supabase for real multi-device use

Hosting

Vercel serves the built website; every push to GitHub redeploys

Quality

100 automated tests, incl. every screen in Hindi and at large text

Live demo: 5 minutes

1

Guided demo with voice

Walk → no alert. Loose band → ignored. Coffee → explained alert. Doctor says false → held back next day. Real problem → urgent alert. Oxygen 89% → safety floor.

2

Doctor

Dashboard → Patients → Jai → note and prescription → Schedule

3

Nurse

Tick a shift task, write a handover, open the ward

4

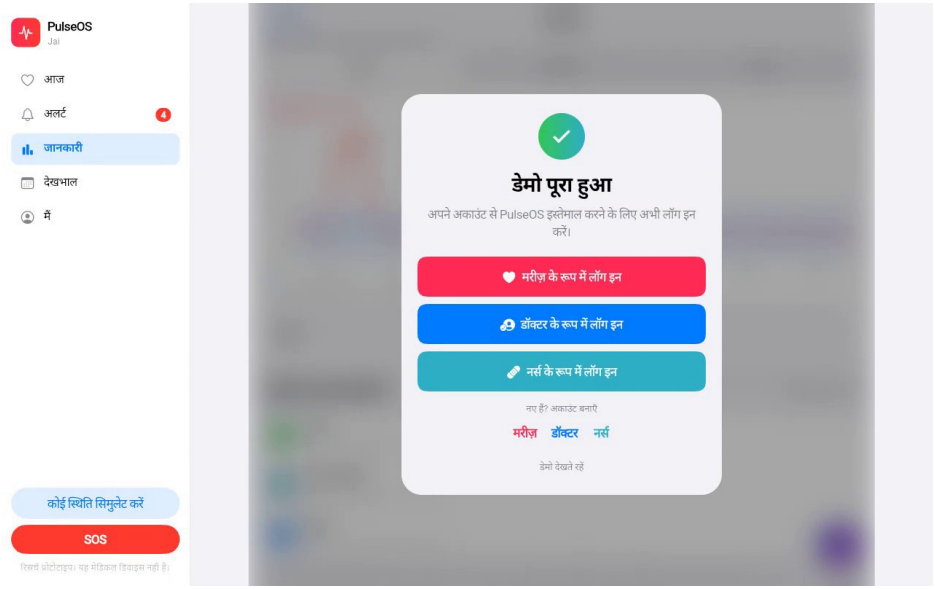
Patient

Help: “report kahan hai?” · Me › Accessibility: easy home

5

Research Lab

Run the experiment, show the chart and table



The guided demo ends by asking you to log in as a patient, doctor or nurse.

Limitations and next steps

Honest limitations

- Results come from a simulation with assumed event rates, not real patients.
- The band in the demo is simulated.
- The decision tree is hand-built, not trained on data.
- Hindi and accessibility need a user study with older and disabled people.

Real wrist band

ESP32 sensors or Apple Watch (HealthKit)

Real data

Replay PhysioNet / MIMIC recordings

Volunteer study

5–10 people for a week, before vs after

Shared backend

Supabase so patients and doctors use different devices

Alerts people can trust again

PulseOS learns each patient, explains every alert and never ignores danger. In simulation: ~97% fewer false alarms and every real event caught.

Try it: the PulseOS website · demo logins doctor@pulseos.app / doctor123
Code: github.com/NANDINISHARMA04/bt-39

Thank you · Questions?